ELSEVIER

Applied Acoustics 61 (2000) 441–453

applied acoustics

www.elsevier.com/locate/apacoust

The effectiveness of floating slab track system — Part I. Receptance methods

F. Cui [a,*], C.H. Chew [b]

[a] *Institute of High Performance Computing, 89C Science Park Drive, The Rutherford, Singapore 118261, Singapore*

[b] *Department of Mechanical and Production Engineering, National University of Singapore, 10 Kent Ridge Crescent, Singapore 119260, Singapore*

Received 22 June 1999; received in revised form 5 November 1999; accepted 10 February 2000

Abstract

The effectiveness of a floating slab track system to stationary harmonic loads and moving harmonic loads are investigated in this paper by using the receptance method.The results show that the floating slab track system is effective in reducing the forces transmitted at frequencies above the designed frequency compared to the fixed slab track system. © 2000 Elsevier Science Ltd. All rights reserved.

1. Introduction

Three types of track system are used in the Singapore Mass Rapid Transit (SMRT) system. The conventional timber sleepers on stone ballast are used on the viaduct route. Concrete sleepers set in a continuous concrete slab (here referred to as fixed slab type or FT for short, see Fig. 1) are used in tunnels, and floating slab track (FST for short, see Fig. 2) which consists of discrete concrete units supported by resilient pads are used in tunnels through sensitive areas where groundborne vibration may be troublesome to the industrial or/and commercial activities.

The floating slab track is designed to reduce the vibration transmitted to the ground. A single degree of freedom analysis is initially used to analyze the floating slab track system. A maximum deflection of 3 mm under the static load of the train is generally imposed to limit the rail deflection. The mass of the floating slab was designed to exceed the average weight of the train in order to avoid modal interactions

* Corresponding author.

0003-682X/00/$ - see front matter © 2000 Elsevier Science Ltd. All rights reserved.
PII: S0003-682X(00)00014-1

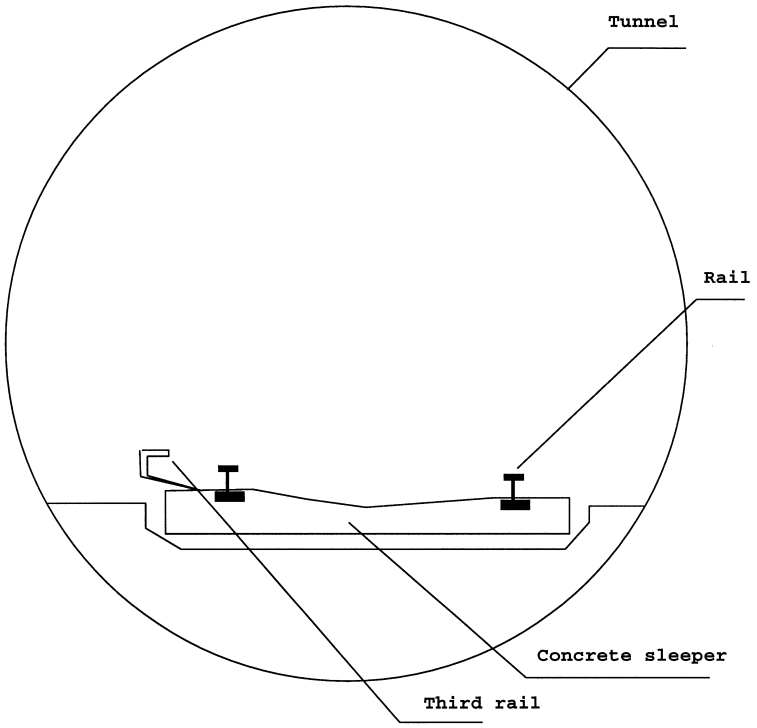


Fig. 1. Fixed slab track.

and to limit the amount of movement of the FST. The natural frequency of the floating slab system is designed to be in the region of 10 Hz. After aging this frequency would be expected to increase to 12~24 Hz.

There is considerable published work on railway track dynamics. The work carried out by Grassie et al. [1–4], Duffy [5], Thompson [6] and others provides us with a comprehensive understanding of track dynamic behaviour. A continuous beam model has been used in modelling groundborne vibration [7], while for wheel/rail interaction noise and high frequency vibration analysis, the discrete supports of sleepers have to be taken into consideration [6]. Wilson et al. [8] introduced the design and test for the continuous and discrete floating slab. As the single degree freedom or two degree freedom models are oversimplified, the effects of changes in the stiffness and damping of railpad and slab elastomer bearing cannot be deduced.

The parameters for the Singapore MRT train are as listed in Table 1. The values of the parameters may vary a little along the line.

2. Continuous models due to moving harmonic loads

In the literature [9], several models have been developed to illustrate the interaction of wheel/rail system. The models chosen to analyze the response of track

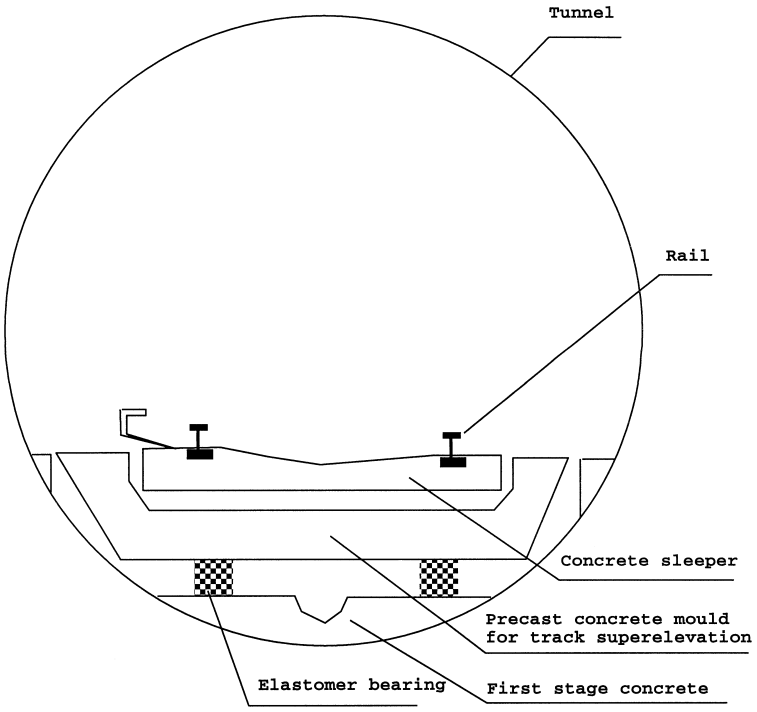


Fig. 2. Floating slab track.

Table 1
Parameters used in the model

Car empty (kg)	30 400
Suspended mass (kg)	2058
Unsuspended mass of axle (kg)	2512
Mass of rail m (kg/m)	60.34
Cross section area A (m^2)	0.7686×10^{-2}
Sleeper l (m)	0.7
Moment of Young's modulus E (GPA)	200
Moment of inertia I (m^4)	3.05×10^{-5}
Timoshenko shear coefficient K	0.34
Bending rigidity EI (MN m^2)	6.1
Shear rigidity GAK (MN m^2)	201
Stiffness of rail pad k_p (Mn/m)	100
Damping constant of rail pad c_p (Ns/m)	30 000
Mass of floating slab m_s (kg/m)	3280
Stiffness of elastomeric bearing k_b (KN/mm)	7.31
Damping constant of elastomeric bearing c_b (Ns/m)	35 000

depend on the frequency range, vibrational direction, and the precision of the results required. It was shown that for moderate frequency excitation a continuous model is accurate enough. The continuous models for the FT and FST systems are as shown in Figs. 3 and 4, respectively. A little difference of rail receptance can be observed both for amplitude and phase lag if the rail is modelled as an Euler beam rather than a Timoshenko beam, as the shear makes the rail more flexible [1]. Here the rail is represented by an infinite Timoshenko beam for general consideration. Both the rail pads and the elastomer bearings which support the floating slab are represented by uniform damped elastic layers. The slab is treated as a lumped mass m_s.The stiffness and viscous damping constant per unit length for rail pads and elastomer bearings k_p^*, c_p^*, k_b^*, c_b^* are obtained by dividing k_k, c_p, k_b, c_b with the sleeper spacing l (k_p, c_p, k_b, c_b are the stiffness of railpad, damping of railpad, stiffness of elastomer bearing, damping of elastomer bearing, respectively).

2.1. Fixed slab track

The equations of motion are (for a Timoshenko beam):

$$GAK\frac{\partial}{\partial x}\left(\frac{\partial y}{\partial x}-\psi\right)+P\mathrm{e}^{j\omega t}\delta(x-vt)-k_p^*y-c_p^*\frac{\partial y}{\partial t}$$
$$=m\frac{\partial^2 y}{\partial t^2}EI\frac{\partial^2\psi}{\partial x^2}+GAK\left(\frac{\partial y}{\partial x}-\psi\right)=\rho I\frac{\partial^2\psi}{\partial t^2} \tag{1}$$

where P is the amplitude of moving load. The receptance of the FT system can be obtained by performing a Laplace transformation on the above equations, giving

$$\alpha(s)=Y(s)/P=\frac{A+Bs^2}{Cs^4+Ds^2+H}=N(s)/D(s) \tag{2}$$

where S is the Laplace transform variable, $A=L\left(\rho Iq^2L^2+GAKL^2\right)$, $B=-LEI$, $C=GAKEI$, $D=-mq^2L^2EI-GAK\rho Iq^2L^2-k_p^*L^2EI-c_p^*qL^2EI$, $H=mq^4L^4\rho I+mq^2L^4GAK+k_p^*L^4\rho Iq^2+k_p^*L^4GAK+c_p^*L^4\rho Iq^3+c_p^*qL^4GAK$ and $q=\omega(j-vs)$.

The receptance can be now found by evaluating the integral

$$\alpha(\xi)=\frac{1}{2\pi j}\int_{\gamma-j\infty}^{\gamma+j\infty}\alpha(s)\mathrm{e}^{s\xi}\mathrm{d}s \tag{3}$$

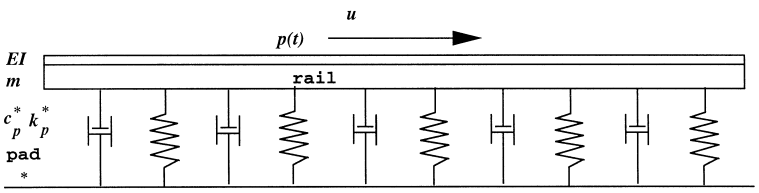


Fig. 3. Continuous model for FT system.

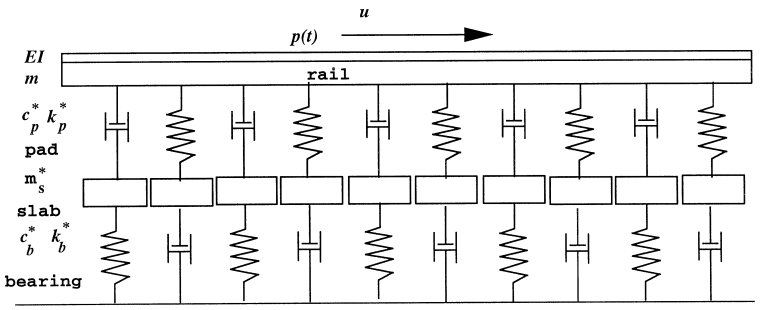


Fig. 4. Continuous model for FST system.

where $\xi = \frac{x-vt}{(EI/\beta)^{1/4}}$. The direct receptance is

$$\alpha(0) = \Sigma_k \mathrm{Res}\{\alpha(s_k)\} \tag{4}$$

where $\mathrm{Res}\{\alpha(s_k)\}$ is all the residues within the contour integral and $Re(s) < 0$.

In order to obtain the residues of Eq. (3) for the case of a moving load, Grassie [1] first obtained the stationary poles (for $v = 0$). Using these values as initial values, the poles for moving load case can be obtained by numerical techniques. Here a function called RESIDUE of MATLAB [10] is used to obtain all the residues. Fig. 5 gives the receptance of a fixed slab track to a stationary and moving load with velocity of 20 and 100 m/s. It can be seen that a moving load with a velocity as high as 100 m/s has very little effect on the receptance.

2.2. Floating slab track

The equations of motion for a moving load with constant velocity v for rail and floating slab are as follows

$$m\frac{\partial^2 y}{\partial t^2} + GAK\frac{\partial}{\partial x} = P\mathrm{e}^{j\omega t}\delta(x - vt) - f(x, t) \tag{5}$$

$$\rho I\frac{\partial^2 \psi}{\partial t^2} - EI\frac{\partial^2 \psi}{\partial x^2} + GAK\left(\psi - \frac{\partial y}{\partial x}\right) = 0 \tag{6}$$

and

$$m_s\frac{\partial^2 w}{\partial t^2} = f(x, t) - g(x, t) \tag{7}$$

$$f(x, t) = \left(k_p^* + c_p^*\frac{\partial}{\partial t}\right)(y - w) \tag{8}$$

$$g(x,t) = \left(k_b^* + c_b^* \frac{\partial}{\partial t}\right) w \tag{9}$$

w is the displacement of the floating slab and $f(x, t)$ and $g(x, t)$ are the comprehensive forces per unit length in the rail pad and elastomer bearing, respectively.

By performing a Laplace transformation to the above rail and slab equations, the receptance for the floating slab track is obtained

$$\alpha(s) = \frac{b_2 c_3 / L}{a_1 b_2 c_3 + b_1^2 c_3 - b_2 c_1^2} = \frac{N(s)}{D(s)} \tag{10}$$

where $a_1 = mq^2 - GAK/L^2 s^2 + k_p^* + c_p^* q$, $b_1 = GAKs/L$, $c_1 = (k_p^* + c_p^* q)$, $b_2 = \rho I q^2 + GAK - EI/L^2 s^2$, $c_3 = m_s q^2 + k_p^* + c_p^* q + k_b^* + c_b q$, $L = (EI/k^*)^{1/4}$, $k^* = k_p^* k_b^* / (k_p^* + k_b^*)$.

The numerator and the denominator of Eq. (10) are funtions of s with highest order of 4 and 6, respectively. Again the receptance can be obtained by a similar procedure as described for FT system. Fig. 6 gives the receptance for both a stationary and moving train with velocity up to 100 m/s. Two resonance frequencies can be identified: one at 8 Hz and the other at 232 Hz. It can be found that the maximum receptance of the FST system at about 8 Hz is 20 times higher than that of an FT System. The roughness amplitude varies from 0.75 to 25 μm in the 1.3 to 30 cm wavelength range [11]. As the speed of the Singapore MRT tain is around 20 m/s, the frequency of roughness excitation will be in the range of 66 to 1538 Hz ($v = \lambda f$). At this frequency range, the receptance of the FST system is comparable to that of an FT system. No system stability problem would be caused by the FST system.

2.3. Forces transmitted to the ground

The force transmitted to the ground for an FT system is given by

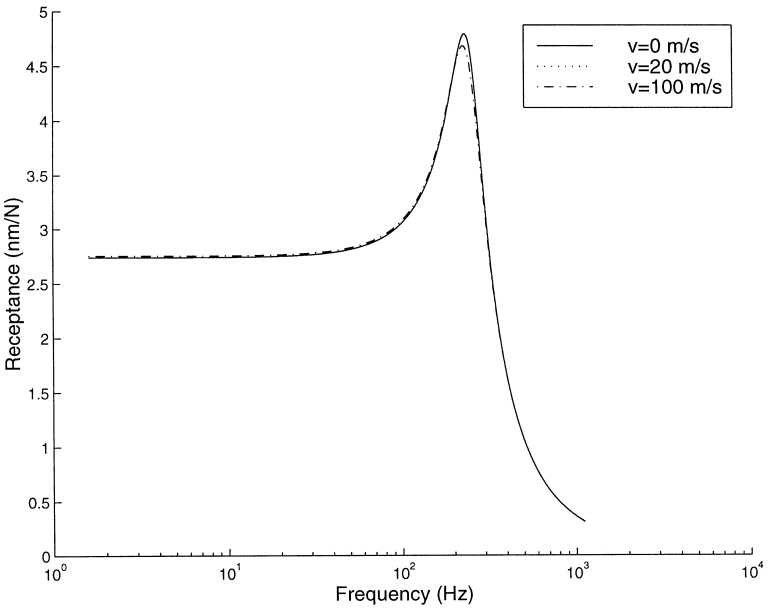
$$R_{FT}(s) = \left(k_p^* + c_p^* q\right) \frac{N(s)}{D(s)} \tag{11}$$

The transmitted force for unit moving harmonic load is shown in Fig. 7a. The force transmitted to the ground for an FST system which is given by

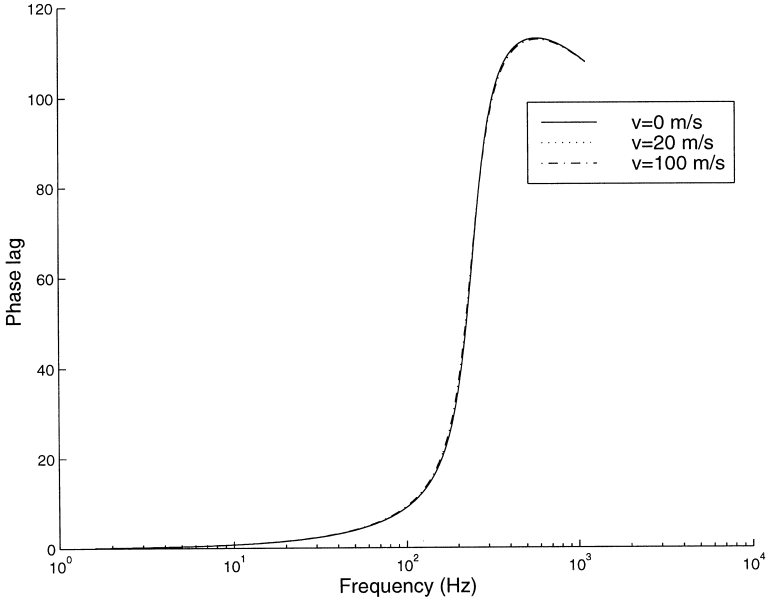
$$R_{FST}(s) = \frac{\left(k_b^* + c_b^* q\right)\left(k_p^* + c_p^* q\right)}{m_s q^2 + k_p^* + c_p^* q + k_b^* + c_b^* q} \alpha(s) \tag{12}$$

is shown in Fig. 7b.

From Fig. 7, it can be seen clearly that a floating slab track can effectively reduce the dynamic forces transmitted to the ground for the excitation frequency of moving load larger than 1.5 times of the designed resonace frequency of FST (above 15 Hz).

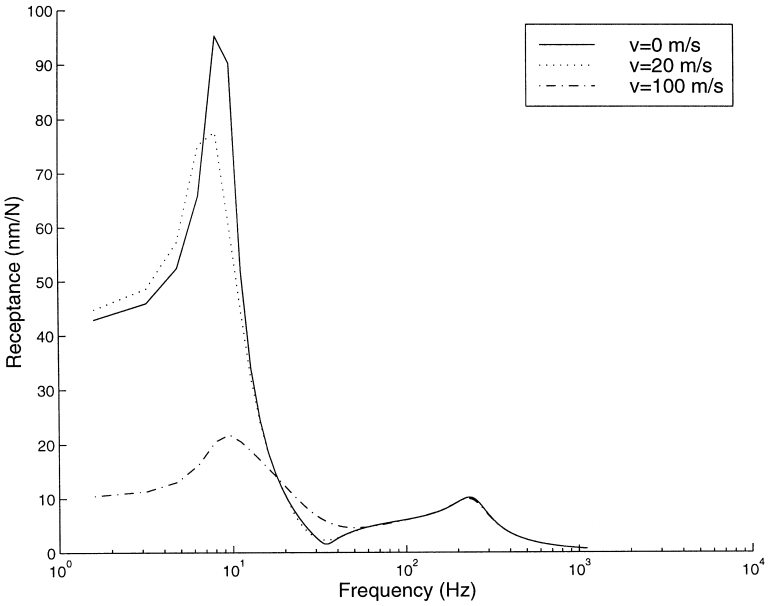


a) Amplitude

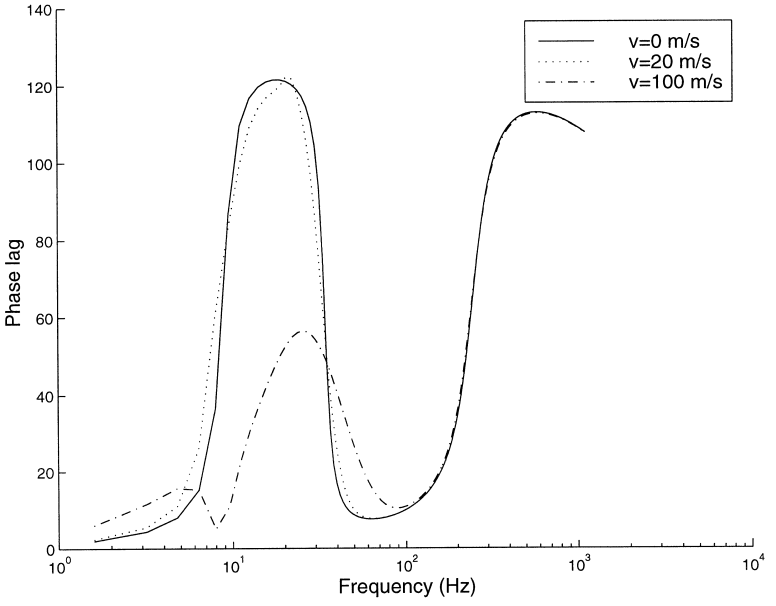


b) Phase lag

Fig. 5. Receptance for FT system: (a) amplitude; (b) phase lag.

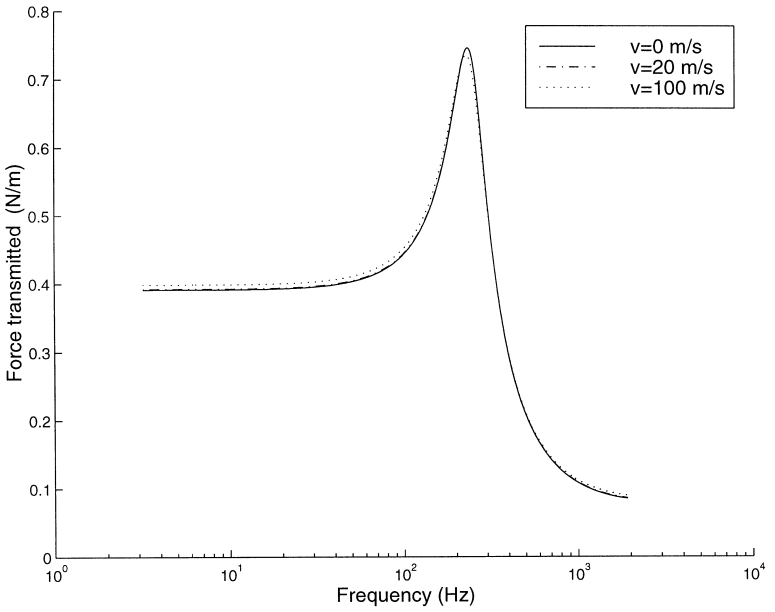


a) Amplitudes

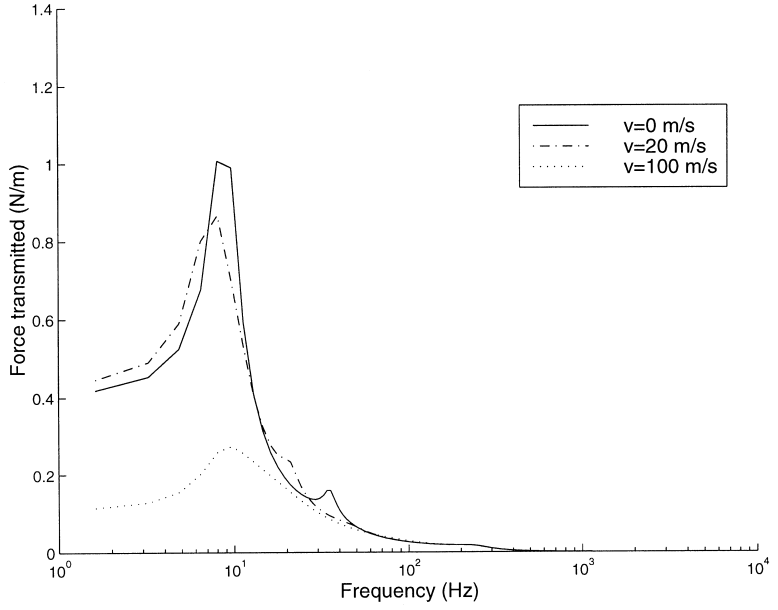


b) Phase lag

Fig. 6. Receptance for FST system: (a) amplitude; (b) phase lag.



a) FT system



b) FST system

Fig. 7. Forces transmitted to the ground: (a) FT system; (b) FST system.

At 8 Hz, the maximum force transmitted for an FST is higher than the case of an FT system due to the heavy weight of the slab mass. However, the frequency corresponding to the maximum force is below the designed frequency already.

3. Discrete models due to stationary loads

The discrete models for FT and FST systems are depicted in Figs. 8 and 9, respectively. The transfer matrix method [1] is used to obtain the receptance of FT and FST systems. The receptance and force transmitted for the FT and FST are shown in Figs. 10 and 11, respectively (all results refer to left support point. a is the distance between the load and left support point).

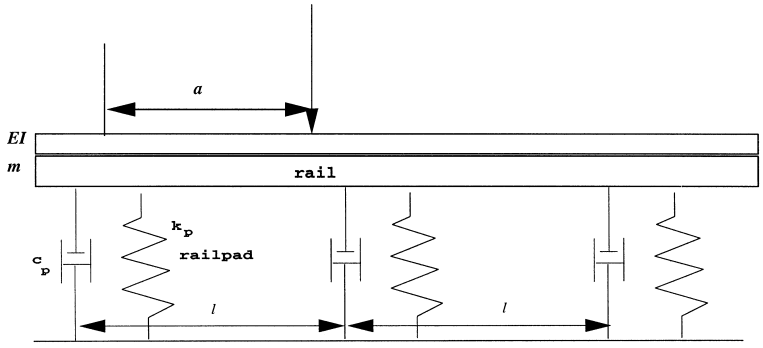


Fig. 8. Discrete model for FT system.

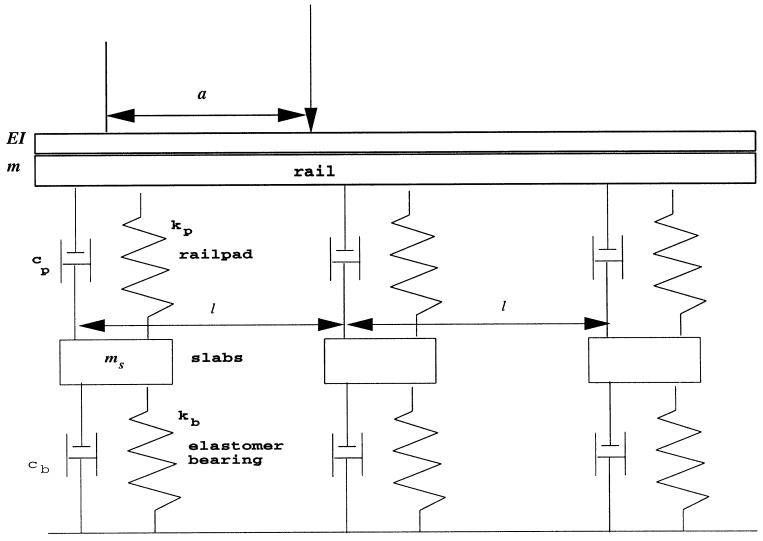
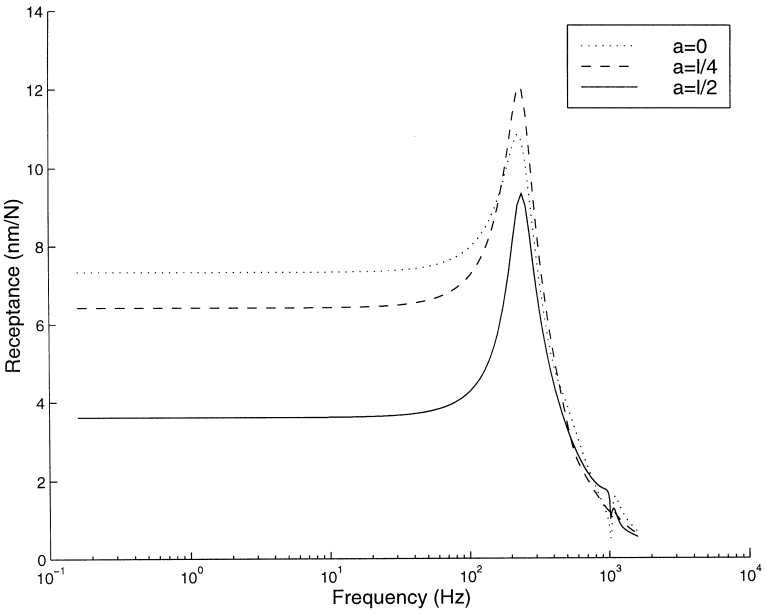
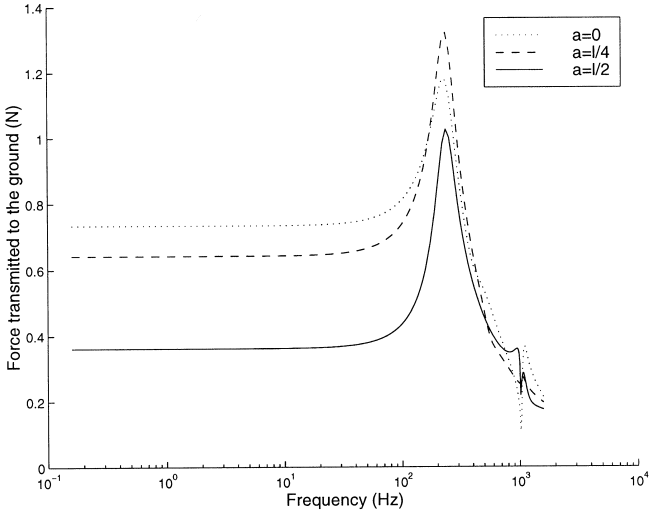


Fig. 9. Discrete model for FST system.

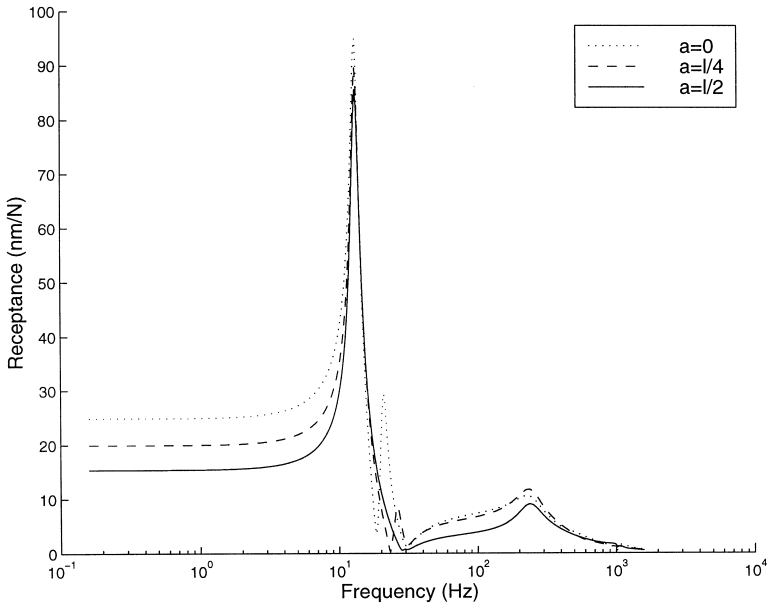


a) Receptance

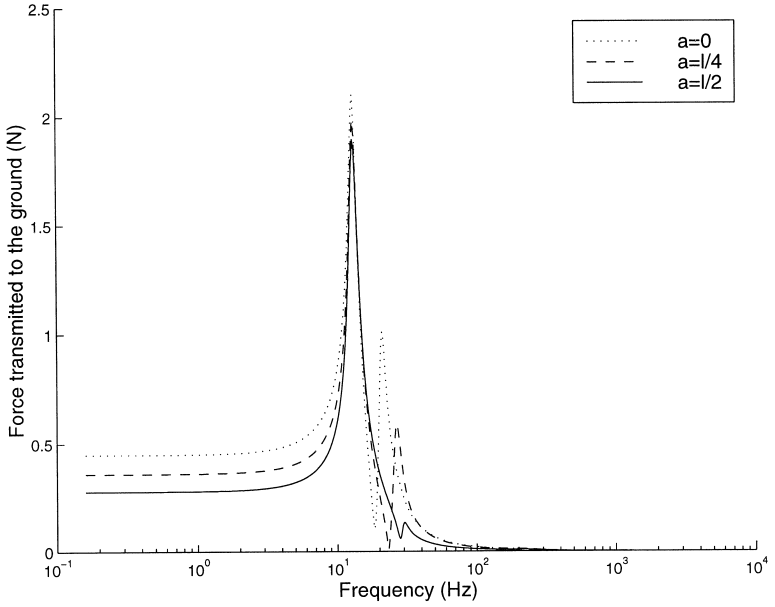


b) Force transmitted to the ground.

Fig. 10. FT system using discrete model: (a) receptance; (b) force transmitted to the ground.



a) Receptance



b) Force transmitted to the ground.

Fig. 11. FST system using discrete model: (a) receptance; (b) force transmitted to the ground.

The general trend for receptances from the discrete models is similar to that obtained from the continuous models. However, a resonance region 1000 Hz is obtained as shown in Fig. 10. This is due to the so-called 'pinned-pinned' resonance, the wavelength of the travelling wave in an Euler beam is equal to twice the sleeper spacing. The resonance frequency is given by

$$\omega_{\mathrm{E}} = (\pi/l)\sqrt{(EI/m)} \tag{13}$$

and is equal to 1019.3 Hz for our track system.

Comparing Fig. 10 with Fig. 11, we find that FST system can effectively reduce the forces transmitted to the ground if the frequency of excitation force is greater than 15 Hz.

4. Conclusion

In this paper, the dynamic responses and forces transmitted to ground of FST and FT systems are investigated. Theoretical results for moving harmonic loads and stationary loads are obtained by using the receptance method. The theoretical results show that for the Singapore FST system the force transmitted to the ground can be reduced greatly for excitation frequencies greater than 15 Hz.

References

[1] Grassie SL, Gregory RW, Harrison D, Johnson KL. The dynamic response of railway track to high frequency vertical excitation. Journal of Mechanical Engineering Science 1982;24(2):77–90.
[2] Grassie SL, Gregory RW, Johnson KL. The dynamic response of railway track to high frequency lateral excitation. Journal of Mechanical Engineering Science 1982;24(2):91–5.
[3] Grassie SL, Gregory RW, Johnson KL. The dynamic response of railway track to high frequency longitudinal excitation. Journal of Mechanical Engineering Science 1982;24(2):97–102.
[4] Grassie SL, Gregory RW, Johnson KL. The behaviour of railway wheelsets and track at high frequencies of excitation. Journal of Mechanical Engineering Science 1982;24(2):103–11.
[5] Duffy DG. The response of an infinite railroad track to a moving, vibrating mass. Trans of the ASME, Journal of Applied Mechanics 1990;57:66–73.
[6] Thompson DJ. Wheel–rail noise generation, part 3: rail vibration. Journal of Sound and Vibration 1993;161(3):421–46.
[7] Kim KJ, Lee JH, Kim Y, Kim MB. Dynamic characteristics of rail track with viscoelastic vibration isolation components. In: Inter Noise-94, Yokohama, Japan, 29–31 August. p. 129–34.
[8] Wilson GP, Saurenman HJ, Nelson JT. Control of ground-borne noise and vibration. Journal of Sound and Vibration 1983;87(2):339–50.
[9] Knothe KL, Grassie SL. Modelling of railway track and vehicle/track interaction at high frequencies. Vehicle System Dynamics 1993;22:209–62.
[10] Palm WJ. MATLAB for engineering applications. Boston: WCB/McGraw–Hill, 1999.
[11] Bender EK, Remington PJ. The influence of rails on train noise. Journal of Sound and Vibration 1974;37:321–34.